

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Emergency Operating Procedures		
TASK:	Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)		
TASK NUMBER:	N1150290501		
JPM NUMBER:	15-01 NRC Sim d		
ALTERNATE PATH:	☒	K/A NUMBER:	EPE E05 EA1.1
APPLICABILITY:	EO ☐	IMPORTANCE FACTOR:	4.1 RO 4.0 SRO
	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	2-EOP-FRHS-1, Loss of Secondary Heat Sink, Rev. 30 31 <i>10/12/16</i>		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>6 minutes</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-2-16
Validated By:	S Harris / J Pierce SME or Instructor	Date:	9-28-16
Approved By:	<i>[Signature]</i> Training Department	Date:	<i>10/11/16</i>
Approved By:	<i>[Signature]</i> Operations Department	Date:	<i>10/11/16</i>
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

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SYSTEM: Emergency Operating Procedures

TASK: TCAF a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

TASK NUMBER: N1150290501

SIMULATOR SETUP: RX HEAD VENT KEYS ARE LOCATED IN SIM BOOTH

IC-254 developed by : MSL rupture (10%) downstream of MSIVs. Fail all MSIVs open. All AFW pumps failed. Fail 2PR2 shut. Performed TRIP-1 through Step 20. SG WR levels <32%.

INITIAL CONDITIONS:

- Unit 2 initiated a Rx trip from 100% power in response to a Main Steamline break at the mixing bottle.
- An automatic Safety Injection initiated.
- The Main Turbine failed to trip automatically, and was manually tripped from the control console.
- MSLI failed, and all MSIV's remain open.
- All AFW flow has been lost.
- EOP-TRIP-1 was performed and a transition to FRHS-1, Loss of Secondary Heat Sink was made at Step 20.1

INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Stop ALL RCPs.
2. Open 2PR1.
3. Open Rx Head Vent Valves 2RC40-2RC43.

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TASK: Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	1	IS TOTAL AFW FLOW LESS THAN 22E04 LB/HR DUE TO OPERATOR ACTIONS	Determines it was not operator action which caused total AFW flow to be less than 22E04 lb/hr.		
	2	<u>IF AT LEAST ONE INTACT OR RUPTURED SG IS AVAILABLE, THEN DO NOT FEED A FAULTED SG</u>	Recognizes ALL SGs are faulted.		
	3	IS RCS PRESSURE GREATER THAN <u>ANY INTACT OR RUPTURED SG PRESSURE</u>	Checks RCS pressure on control console and determines it is greater than all SG pressures checked on control console.		
	3.1	ARE RCS T-HOTS GREATER THAN 350°F	Checks RCS Thot indication on control console and determines that RCS Thots are greater than 350°F.		
	4	<u>IF WR LEVELS IN AT LEAST TWO SG's ARE LESS THAN 32% (37% ADVERSE), THEN IMMEDIATELY GO TO STEP 21 TO INITIATE RCS BLEED AND FEED</u>	Determines at least 2 SG WR levels < 32% and goes to Step 21.		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	21	<u>CAUTION</u> TO ESTABLISH RCS HEAT REMOVAL BY RCS BLEED AND FEED, STEPS 24 THRU 29 MUST BE PERFORMED QUICKLY AND WITHOUT INTERRUPTION	Reads Step.		
*	23	STOP <u>ALL</u> RCPS	Depresses STOP PB for 21-24 RCPs and verifies green stop light illuminates and red start light extinguishes.		
	22	INITIATE SI	Uses Safeguards key and initiates SI on at least one train of Safeguards initiation.		
	23	ARE SI VALVES IN SAFEGUARDS POSITION	Checks 2RP4 and/or console indication to determine that all valves listed in Table B are in Safeguards position. <u>Table B valves are:</u> 2SJ4 OPEN BIT INLET 2SJ5 OPEN BIT INLET 2SJ12 OPEN BIT OUTLET 2SJ13 OPEN BIT OUTLET 2CV68 CLOSED CHARGING DISCHARGE 2CV69 CLOSED CHARGING DISCHARGE (continued next page)		

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*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			<u>Table B Valves (con't):</u> 21-24SJ54 OPEN ACCUMULATOR OUTLET 2SJ1 OPEN RWST TO CHARGING 2SJ2 OPEN RWST TO CHARGING 2CV40 CLOSED DISCHARGE STOP 2CV41 CLOSED DISCHARGE STOP		
	23.1	IS 21 OR 22 CHARGING PUMP RUNNING	Checks control console and determines 21 and 22 charging pump are running.		
		IS BIT FLOW ESTABLISHED	Checks console and determines BIT flow is established.		
	23.2	IS <u>ANY</u> SI PUMP RUNNING	Checks control console and determines 21 and 22 SI pumps are running.		
		ARE VALVES IN TABLE C OPEN FOR <u>AT LEAST</u> ONE RUNNING SI PUMP	Checks control console indication for valves listed in Table C and determines the valves are open for at least one running SI pump. <u>Table C valves are:</u> <u>21/22 SI PUMPS</u> 2SJ30 (FROM RWST) 21/22SJ33 (SI PUMP SUCTION) 2SJ135 (COLD LEG DISCHARGE) 21/22SJ134 (COLD LEG DISCHARGE)		

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SYSTEM: **Emergency Operating Procedures**

TASK: **Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		ARE ALL CHARGING AND SI PUMPS RUNNING AND ALIGNED TO DELIVER FLOW	Determines all charging and SI pumps are running and aligned to deliver flow.		
	24	OPEN <u>BOTH</u> PZR PORV STOP VALVES	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		
*		OPEN <u>BOTH</u> PZR PORVS	Places 2PR1 in manual and depresses 2PR1 PZR PORV open PB and verifies green closed light extinguishes and red open light illuminates. Places 2PR2 in manual and depresses 2PR2 PZR PORV open PB and reports that 2PR2 will not open.		
	24.1	ARE <u>BOTH</u> PZR PORV STOP VALVES OPEN	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		
		ARE <u>BOTH</u> PZR PORVS OPEN	Determines 2PR2 PZR PORV is not open. Alternate path actions start here.		
*	24.2	OPEN 2RC40 THRU 2RC43 (REACTOR HEAD VENTS)	Inserts key into each 2RC40 THRU 2RC43 (REACTOR HEAD VENTS) switch on 2RP3, turns to open, and verifies each valve opens.		
			Terminate JPM when operator has opened 2RC40 thru 2RC43 Reactor Head Vents.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- u ✓ 2 h 1. Task description and number, JPM description and number are identified.
- u ✓ 2 h 2. Knowledge and Abilities (K/A) references are included.
- u ✓ 2 h 3. Performance location specified. (in-plant, control room, or simulator)
- u ✓ 2 h 4. Initial setup conditions are identified.
- u ✓ 2 h 5. Initiating and terminating Cues are properly identified.
- u ✓ 2 h 6. Task standards identified and verified by SME review.
- u ✓ 2 h 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 2 h 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 30 Date 7/28/16
- 2 h 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: J. PIERCE JP

Date: 7/28/16

SME/Instructor: S. Harris SH

Date: 9/28/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

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- An automatic Safety Injection initiated.
- The Main Turbine failed to trip automatically, and was manually tripped from the control console.
- MSLI failed, and all MSIV's remain open.
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INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.